

Recitation-1

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BSCS - 3A

CS311
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Code

```
1  using System;
2  using System.Collections.ObjectModel;
3  using System.ComponentModel.DataAnnotations;
4  using CommunityToolkit.Mvvm.ComponentModel;
5  using CommunityToolkit.Mvvm.Input;
6
7  namespace ViewModels;
8
9  public partial class MainViewModel : ObservableValidator
10 {
11
12     [ObservableProperty]
13     [Required(ErrorMessage = "Input 1 is required.")]
14     [CustomValidation(typeof(MainViewModel), nameof(ValidateNumber))]
15     private string? _input1 = "";
16
17     [ObservableProperty]
18     [Required(ErrorMessage = "Input 2 is required.")]
19     [CustomValidation(typeof(MainViewModel), nameof(ValidateNumber))]
20     private string? _input2 = "";
21
22     [ObservableProperty]
23     private string? _result;
24
25     public static ValidationResult? ValidateNumber(string? value,
26         ValidationContext context)
27     {
28         if (double.TryParse(value, out _))
29             return ValidationResult.Success;
30
31         return new ValidationResult("Must be a valid number.");
32     }
33
34     [RelayCommand]
35     private void Add()
36     {
37         ValidateAllProperties();
38         if (HasErrors)
```

```
39         Result = "Invalid input.";
40         return;
41     }
42
43     if (double.TryParse(Input1, out double num1) && double.TryParse(Input2,
44         out double num2))
45     {
46         Result = (num1 + num2).ToString("0.0");
47     }
48 }
49
50 [RelayCommand]
51 private void Subtract()
52 {
53     if (double.TryParse(Input1, out double num1) && double.TryParse(Input2,
54         out double num2))
55     {
56         Result = (num1 - num2).ToString("0.0");
57     }
58 }
59
60 [RelayCommand]
61 private void Multiply()
62 {
63     if (double.TryParse(Input1, out double num1) && double.TryParse(Input2,
64         out double num2))
65     {
66         Result = (num1 * num2).ToString("0.0");
67     }
68 }
69 [RelayCommand]
70 private void Divide()
71 {
72
73     if (double.TryParse(Input1, out double num1) && double.TryParse(Input2,
74         out double num2))
75     {
76         if (num2  $\neq$  0)
77         {
78             Result = (num1 / num2).ToString("0.0");
79         }
80     }
```

Screenshots

- Both inputs are empty and click subtract.

A screenshot of a calculator application on a dark background. It features two input fields labeled "Input number:". The first input field is empty, and below it, the text "Input 1 is required." is displayed in red. The second input field is also empty, with "Input 2 is required." displayed in red below it. Between the input fields is a row of four buttons: "Add", "Subtract", "Multiply", and "Divide". At the bottom, there is a "Result:" label followed by an empty output field.

- Input 1 is empty and input2 is a character/string and click subtract.

A screenshot of the calculator application. The first input field is empty with the error "Input 1 is required." below it. The second input field contains the text "foo", and below it, the error "Must be a valid number." is displayed in red. The buttons "Add", "Subtract", "Multiply", and "Divide" are visible. The "Result:" field at the bottom is empty.

- Input 1 is a character/string and input 2 is empty and click subtract.

A screenshot of the calculator application. The first input field contains the text "fod", and below it, the error "Must be a valid number." is displayed in red. The second input field is empty with the error "Input 2 is required." below it. The buttons "Add", "Subtract", "Multiply", and "Divide" are visible. The "Result:" field at the bottom is empty.

- Both inputs are character and string and click subtract.

Input number:
Must be a valid number.

Input number:
Must be a valid number.

Result:

- Input 1 is empty and input 2 is a number and click subtract.

Input number:
Must be a valid number.

Input number:

Result:

- Input 1 is a number and input 2 is empty and click subtract.

Input number:

Input number:
Input 2 is required.

Result:

- Input 1 is a character/string and input 2 is a number and click subtract.

Input number:
Must be a valid number.

Input number:

Result:

- Input 1 is a number and input 2 is a character/string and click subtract.

Input number: 34

Input number: bar

Must be a valid number.

Add Subtract Multiply Divide

Result:

- Both inputs are number and click subtract.

Input number: 72

Input number: 12

Add Subtract Multiply Divide

Result: 60.0

- Both inputs are number and click multiply.

Input number: 72

Input number: 12

Add Subtract Multiply Divide

Result: 864.0

- Both inputs are number and click divide.

Input number: 72

Input number: 12

Add Subtract Multiply Divide

Result: 6.0

<https://drive.google.com/file/d/1Ydo-H7juLxrJCKtv3INWafdiFmhECI7l/view?usp=sharing>